Projecting the Healthcare Affordability Gap:
A Stochastic Stress-Test of Out-of-Pocket Healthcare Burden for Ghana's Generation Z Cohort, 2026–2061

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Abstract

Ghana's National Health Insurance Scheme (NHIS) was designed to shield households from catastrophic out-of-pocket (OOP) healthcare costs, yet OOP payments still account for approximately 26 percent of current health expenditure nationally — well above the ceiling the World Health Organization recommends (Okoroh and Essoun 2018). This paper asks a generational question that the largely retrospective catastrophic-health-expenditure (CHE) literature does not directly address: if current trends in healthcare cost growth, wage growth, and macroeconomic volatility persist, what OOP burden will Ghana's Generation Z cohort face by the age at which today's CHE literature is typically anchored — roughly 50? We combine two complementary approaches. First, we compute household OOP burden directly from GLSS7 (2016/17) microdata (n = 14,009 households) using an original extraction pipeline, finding a weighted mean burden of 0.75 percent of household expenditure, rising to 0.92–1.04 percent among households aged 60 and older, and a modest but directionally consistent gap by household-head literacy status (0.74 percent for literate-headed households versus 0.84 percent for illiterate-headed households). Second, we construct a stylized Monte Carlo stress-test simulation, calibrated to Ghana's minimum-wage growth series, historical and recent consumer price inflation, national health-financing shares, and CHE incidence from the wider literature — with every parameter classified as Ghana-specific empirical, external empirical, or author-calibrated (Section 3.1) — simulating 10,000 household trajectories under three scenarios: a status-quo continuation of current trends; a financing-reform scenario reaching the Abuja Declaration's 15-percent-of-budget target; and a shock-heavy scenario reflecting Ghana's history of recurrent currency crises. The primary simulation spans 2026–2051 (approximately age 25–50), matching the age at which the CHE literature this paper builds on is typically anchored; under the status-quo scenario, mean household OOP burden rises from approximately 8.1 percent of income in 2026 to 19.6 percent by 2051, and only the financing-reform scenario keeps burden low throughout. We additionally report an extended simulation — the identical model and parameters re-run through 2061 (approximately age 60) — under which status-quo burden reaches 26.7 percent of income and the shock-heavy scenario approaches the entirety of household income; this extension is reported separately from the primary analysis and should not be read as the paper's original design horizon. Because the simulation's catastrophic-event hazard parameter (8 percent annually) was calibrated from wider regional literature rather than this paper's own

GLSS7 finding, we additionally back-solve and re-run the primary and extended simulations under a Ghana-empirical calibration anchored to the 1.08 percent estimate; the paper's scenario ordering and policy conclusions are preserved under this alternative, though absolute projected burden is somewhat lower. A one-at-a-time sensitivity analysis of the primary simulation shows the projection is most sensitive to parameters with the strongest empirical grounding (wage growth, currency-shock frequency) and least sensitive to its more speculative assumptions, though we note this design does not capture interaction effects between parameters and report it as a lower bound on total outcome uncertainty. We discuss the structural drivers behind these trajectories — currency exposure in health financing, the breadth-versus-depth problem in NHIS coverage, the epidemiological transition toward non-communicable disease, and rising informal-sector employment — and the specific policy levers available to alter Generation Z's outcome. Throughout, Generation Z is treated as an analytical cohort rather than a homogeneous population, and all simulation outputs — including values approaching or exceeding household income under the shock-heavy scenario — are reported as model-implied outcomes under stated stress-test assumptions, not deterministic predictions of any individual household's experience; full model equations, the parameter-evidence classification, and simulation code are provided to support independent reproduction.

Keywords: catastrophic health expenditure; out-of-pocket payments; National Health Insurance Scheme; Ghana; generational economic projection; Monte Carlo simulation; universal health coverage; GLSS7 microdata

1. Introduction

A common household narrative in Ghana runs roughly as follows: a person in their late forties or fifties, enrolled in the National Health Insurance Scheme (NHIS), spends a substantial share of household income on routine and unexpected healthcare costs, and a single serious illness — a stroke, a surgery, a chronic diagnosis such as diabetes or hypertension — pushes that household into debt. This pattern is not anecdotal. Ghana's own health-financing data shows that out-of-pocket (OOP) payments remain approximately 26 percent of current health expenditure nationally, more than the 15–20 percent ceiling recommended by the World Health Organization (Okoroh and Essoun 2018), and that catastrophic health expenditure — OOP payments exceeding a defined share of household income or consumption — has a well-documented, regressive incidence that falls hardest on poorer households (Akazili et al. 2017; Ataguba and Kabaniha 2022).

What is less studied is the forward-looking, generational version of this question. Ghana's Generation Z — those born roughly between 1997 and 2012, now entering or newly established in the labor force — will reach the life stage at which catastrophic health expenditure risk concentrates (their late forties and fifties) only after another two to three decades of compounding healthcare cost growth, wage dynamics, and currency volatility. If the structural drivers identified in the retrospective CHE literature continue on their current trajectories, does the healthcare affordability position of Gen Z at age 50 differ meaningfully from that of the cohort reaching age 50 today? And if so, by how much, and through which specific mechanisms?

This paper addresses that question directly. Rather than treating catastrophic health expenditure as a cross-sectional snapshot — the dominant approach in the Ghanaian and Sub-Saharan African literature, which relies on single-round household survey data such as the Ghana Living Standards Survey (GLSS) — we build a forward-projecting, stochastic simulation that compounds Ghana's own historical wage growth, inflation, and health-financing parameters over a 25-year horizon. The approach is deliberately stylized: it is a stress-test in the tradition of World Bank and WHO universal-health-coverage (UHC) financial-protection modeling, not a structural econometric forecast. Its purpose is to make the generational stakes of current health-financing policy legible — to show, under transparent and replicable assumptions, what continuation of the status quo implies for a specific cohort, and how much a credible financing-reform pathway could change that outcome.

The remainder of the paper proceeds as follows. Section 2 reviews the literature on catastrophic health expenditure and health financing in Ghana and the wider Sub-Saharan African region, and situates the paper's contribution relative to that literature. Section 3 describes the data sources, model structure, and scenario design. Section 4 presents results. Section 5 discusses the structural drivers behind the projected trajectories and their generational implications. Section 6 discusses limitations, and Section 7 concludes with policy implications.

2. Literature Review

2.1 Catastrophic health expenditure and NHIS performance in Ghana

Ghana introduced the National Health Insurance Scheme in 2003 with the explicit goal of reducing OOP payments and protecting households from catastrophic and impoverishing health expenditure. The pre-NHIS baseline, established using 2005/2006 Ghana Living Standards Survey (GLSS 5) data, found that OOP healthcare payments increased poverty headcount by 9.4 percent at the $1.25-per-day poverty line, with a substantially regressive incidence across income deciles (Akazili et al. 2017). A companion analysis using the same survey round found that roughly 2.6 percent of households spent more than 20 percent of total household income on OOP healthcare, with the burden concentrated among poorer households even after adjusting for the concentration of spending (Akazili et al. 2017).

Two decades into NHIS implementation, the scheme's effect on financial risk protection remains mixed. A systematic review of NHIS impact studies published between 2003 and 2017 found that uninsured households paid between 1.4 and 10 times more out of pocket than insured households, yet 6 to 18 percent of insured households still incurred catastrophic payments, and all included studies reported OOP spending on medicines even among the insured (Okoroh and Essoun 2018). Ataguba and Kabaniha (2022), recomputing historical Ghanaian data, show that OOP health spending is distributed regressively relative to income: the poorest quintile accounts for 5.5 percent of national income but 6.9 percent of national OOP health spending, while richer quintiles are comparatively under-represented in OOP burden relative to their income share — evidence that ability-to-pay and OOP exposure are not well aligned under the current financing mix. More recent work using later GLSS rounds and the 2022/2023 Annual Household Income and Expenditure Survey documents a declining general incidence of catastrophic expenditure at high consumption thresholds, but persistent and disproportionate risk among the poorest households (Sataru, Twumasi-Ankrah, and Seddoh 2022).

2.2 Structural financing gaps and macroeconomic volatility

Government health spending in Ghana has consistently fallen short of the Abuja Declaration target of 15 percent of the national budget, sitting at approximately 4.2 percent of GDP — below the Sub-Saharan African average (World Bank 2023). Total government health spending nearly doubled in nominal terms between 2015 and 2023 but rose only about 12 percent in real terms, with nearly all of the increase absorbed by wage-bill growth rather than expanded service delivery or NHIS financing depth (World Bank 2023). This financing gap interacts with a second structural feature of the Ghanaian economy: recurrent currency depreciation and inflation shocks. Ghana's inflation rate has ranged from a record low of 0.4 percent (1999) to a record high of 63.1 percent (2001), averaging approximately 17 percent since 1998 (Trading Economics 2026). The most recent shock, in 2022, saw annual inflation exceed 37 percent by official recalculated figures (Bloomberg 2022) and, by some contemporaneous reports, peak above 50 percent before the subsequent disinflation cycle that brought inflation back to single digits by 2025–2026 (Trading Economics 2026). Because a substantial share of Ghana's medical inputs — pharmaceuticals, diagnostic equipment, consumables — are imported and effectively dollar-priced, currency depreciation functions as a de facto healthcare cost shock independent of domestic conditions; Ghana's health expenditure per capita in current US dollars fell 12.8 percent between 2021 and 2022 for exactly this reason (Knoema 2025).

Wage growth has not kept pace with these pressures. Ghana's national daily minimum wage rose from GHS 7.00 in 2015 to GHS 21.77 in 2026 — a nominal increase of roughly 211 percent — but with adjustments that consistently lagged inflation, particularly through the 2015–2022 period, leaving real purchasing power largely stagnant despite nominal gains (Fair Wages and Salaries Commission 2025; Finex Insights 2025). Even after the accelerated wage increases of 2023–2026, the minimum wage remains well below independent living-wage estimates for urban Ghana, which range from GHS 60 to over GHS 100 per day (Playroll 2026).

2.3 Comparative Sub-Saharan African context

Ghana's experience is not unique regionally. Cross-country estimates place the incidence of catastrophic health expenditure across Sub-Saharan Africa between 12.9 and 20.4 percent of households — comparable to South and Southeast Asian middle-income countries such as Bangladesh (14.2 percent) — despite substantial variation in health-financing architecture across the region ("An Analysis of Household Catastrophic Health Expenditure and Food Insecurity in Ghana" 2025). This regional comparability suggests that the mechanisms driving Ghana's trajectory — currency exposure, shallow insurance depth, and wage stagnation relative to healthcare cost inflation — are broadly shared across similarly positioned health systems, which strengthens the external relevance of a Ghana-calibrated model.

2.4 Contribution of this paper

The existing literature on catastrophic health expenditure in Ghana is overwhelmingly retrospective and cross-sectional, built on discrete GLSS survey rounds that capture a single moment in time. This is methodologically appropriate for measuring realized financial catastrophe, but it does not answer a forward-looking, cohort-specific question: given the historical trajectories of the same variables these studies rely on — wage growth, inflation, OOP share, government financing — what does compounding those trends over a full working-life horizon imply for a cohort that has not yet reached the age of peak catastrophic-expenditure risk? This paper's contribution is methodological and generational: it translates retrospective CHE parameters into a forward-projecting stochastic simulation, explicitly designed to make visible the compounding, cumulative nature of a risk that the existing literature measures only in cross-section.

3. Data and Methods

3.1 Data sources

Model parameters are calibrated using the following publicly available sources, each of which is documented in the References section: (i) Ghana's national daily minimum wage series, 2015–2026, anchored to the Government of Ghana's official gazetted rate for 2026 (Fair Wages and Salaries Commission 2025) and a historical compilation for earlier years (Finex Insights 2025); (ii) Ghana consumer price inflation, long-run average and volatility, and recent-year values (Trading Economics 2026; Bloomberg 2022); (iii) out-of-pocket expenditure as a share of current health expenditure, and government health expenditure as a share of GDP (Okoroh and Essoun 2018; World Bank n.d.; World Bank 2023); (iv) current health expenditure per capita in US

dollars, used to characterize currency-driven volatility in health-cost inputs (Knoema 2025); and (v) catastrophic health expenditure incidence and severity estimates from Ghana-specific and Sub-Saharan African studies (Akazili et al. 2017; Sataru, Twumasi-Ankrah, and Seddoh 2022; "An Analysis of Household Catastrophic Health Expenditure and Food Insecurity in Ghana" 2025). Table 1 summarizes the calibrated parameter values and their sources.

Each parameter in Table 1 is classified below according to its evidentiary basis, following the three-tier scheme adopted throughout this paper: (A) Ghana-specific empirical parameter, supported directly by Ghanaian data; (B) external empirical parameter, supported by evidence from other countries or regional studies; and (C) author-calibrated modelling assumption, introduced because Ghana-specific evidence was unavailable or insufficient at the time of calibration. Nominal minimum-wage CAGR, long-run and recent-year CPI inflation, and government health spending as a share of GDP are classified (A), being computed or drawn directly from Ghana-specific series (Fair Wages and Salaries Commission 2025; Trading Economics 2026; Bloomberg 2022; World Bank 2023). The stable- and shock-year inflation moments used to parameterize the model's regime distribution are also classified (A), calibrated to Ghana's own disinflation and 2022–2023 crisis episodes. The annual probability of a shock year is classified (C): it is an author estimate from the observed frequency of Ghanaian currency/inflation crises rather than a published hazard rate. Baseline non-crisis OOP burden and OOP share of current health expenditure are classified (A)/(C) respectively — the latter is drawn directly from Ghana-specific data (Okoroh and Essoun 2018) while the former is an author calibration to the general CHE literature rather than a value estimated directly from GLSS7. The annual catastrophic-event probability (8 percent) is classified (B) in its original form, being blended from Sub-Saharan African literature; Section 3.3, 4.6 and Table 5 report a parallel (A) Ghana-empirical alternative (1.08 percent) back-solved from this paper's own GLSS7 finding, and both are carried through the results transparently rather than treating one as the single correct value. The catastrophic-event cost multiplier (3–7×) is classified (C), an author calibration to the qualitative severity gap documented in the CHE literature rather than a directly estimated distribution. The debt-trigger liquidity buffer and threshold (Section 3.2, component 10) are likewise classified (C) and are discussed further in Section 6.

Parameter	Value	Source
Nominal minimum-wage CAGR, 2015–2026	10.9%	Computed from GHS 7.00 (2015) to GHS 21.77 (2026); Finex Insights 2025; Playroll 2026
Long-run CPI inflation, mean (1998–2026)	17.2%	Trading Economics 2026
Stable-year inflation, simulated mean (sd)	10% (3%)	Author calibration to 2024–2026 disinflation range; Trading Economics 2026
Shock-year inflation, simulated mean (sd)	30% (10%)	Calibrated to 2022–2023 episode; Bloomberg 2022; Trading Economics 2026
Annual probability of a shock year	12%	Author estimate from frequency of Ghanaian currency/inflation crises (≈4 episodes, 2000–2025)
OOP share of current health expenditure	26%	Okoroh and Essoun 2018
Government health spending, % of GDP	4.2%	World Bank 2023
Baseline non-crisis OOP burden, % of income	6%	Author calibration to CHE literature, non-catastrophic years; Akazili et al. 2017
Annual probability of a catastrophic health event	8%	Blended estimate from SSA CHE incidence range (12.9–20.4%) and Ghana-specific declining-incidence findings; "An Analysis of Household Catastrophic Health Expenditure…" 2025; Sataru, Twumasi-Ankrah, and Seddoh 2022

Parameter	Value	Source
Catastrophic-event cost multiplier (of baseline)	3–7×	Author calibration to CHE severity literature

Table 1. Calibrated simulation parameters and data sources.

3.2 Model structure

We simulate n = 10,000 household trajectories over T = 25 annual periods (2026–2051), representing a cohort entering the labor force at approximately age 25 in 2026 and reaching approximately age 50 in 2051 — the life stage at which the Ghanaian CHE literature concentrates its risk estimates. For each household i and year t, the model tracks a nominal income index and a nominal health-cost index, both initialized at 100 in 2026.

Each year, a household is stochastically assigned to a "shock" or "stable" macroeconomic regime, with inflation drawn from a regime-specific normal distribution (Table 1) and clipped to a plausible range of −2 to 65 percent. Income grows at the historical nominal minimum-wage compound annual growth rate, held fixed across regimes to reflect the empirical observation that wage-setting in Ghana has responded only sluggishly to inflation shocks (Fair Wages and Salaries Commission 2025; Finex Insights 2025). Health-cost growth is modeled as the regime inflation rate plus an "import-exposure premium" that is larger in shock years, capturing the empirically observed additional pass-through of currency depreciation into imported medical inputs (Knoema 2025).

The ratio of the cumulative health-cost index to the cumulative income index — the "affordability gap ratio" — scales a household's baseline non-crisis OOP burden in each year. Independently, each household faces an annual probability of a catastrophic health event; in such years, OOP burden is multiplied by a random severity factor (uniform between 3 and 7), consistent with the empirical gap between routine and catastrophic OOP payment levels documented in the CHE literature (Akazili et al. 2017). A household is recorded as experiencing a debt-triggering event when a catastrophic year's OOP burden exceeds an assumed one-month liquid-savings buffer by more than 15 percentage points of annual income.

Throughout this paper, we refer to this T = 25, 2026–2051 model as the PRIMARY SIMULATION. Its endpoint (age 50) was chosen to match the age at which the retrospective catastrophic-health-expenditure literature this paper builds on typically concentrates its risk estimates (Section 2), and all headline results, tables, and figures in Sections 4.1 through 4.3 and 4.5–4.6 refer to this primary simulation unless stated otherwise. Section 4.4 separately reports an EXTENDED SIMULATION: the identical model, parameters, and scenario definitions re-run for an additional 10 annual periods (T = 35, 2026–2061), extending the same cohort's trajectory to approximately age 60. The extended simulation is not a separate or independently calibrated model — it uses exactly the parameter values in Table 1 and Table 2 — but is reported separately because it was not part of the paper's original 25-year design and its results should not be read as having the same status as the primary, age-50-anchored analysis.

Mathematical formulation. This subsection specifies every component of the model precisely enough for independent replication. Each household i is simulated for t = 0, 1, ..., T−1 annual periods, with all index variables initialized to 1.0 at t = 0 (representing the 2026 baseline).

(1) Income growth. Household nominal income follows a deterministic compound growth path fixed at Ghana's historical minimum-wage compound annual growth rate (Table 1), identical across scenarios and households: $I_t = I_{t-1} \times (1 + g_w)$, where g_w = 0.109 (10.9 percent) for all t. Income growth does not vary stochastically or by scenario in this version of the model.

(2) Healthcare-cost growth. The health-cost index compounds each year at the sum of that year's realized inflation rate and an import-cost premium: $H_t = H_{t-1} \times (1 + \pi_t + \delta_t)$, where π_t is inflation (component 3) and δ_t is the import-cost premium (component 5).

(3) Inflation. Each household-year draws an independent macroeconomic regime, $Shock_{it}$ ~ Bernoulli(p_shock_year), where p_shock_year is scenario-specific (Table 2). Conditional on the regime, inflation is normally distributed: π_{it} ~ N(μ_shock, σ²_shock) if $Shock_{it} = 1$, or π_{it} ~ N(μ_stable, σ²_stable) if $Shock_{it} = 0$, with regime means and standard deviations given in Table 1. The realized draw is clipped to the plausible range [−2%, 65%].

(4) Currency/economic shocks. Modeled as the binary regime indicator $Shock_{it}$ defined in component (3). Shocks are drawn independently across households and across years (no serial correlation or shock persistence is modeled within a household's trajectory, and no cross-household correlation is modeled — i.e., the model does not represent a single national shock hitting all households simultaneously in a given year, but rather independent household-level exposure to a shared annual shock probability).

(5) Import-cost exposure. A deterministic premium applied on top of inflation, taking one of two scenario-specific values depending on the realized regime: $\delta_{it} = \delta_shock$ if $Shock_{it} = 1$, or $\delta_{it} = \delta_stable$ if $Shock_{it} = 0$ (values in Table 1/Table 2).

(6) Catastrophic events. Independent of the macroeconomic regime, each household-year draws $Crisis_{it}$ ~ Bernoulli(p_health_crisis), where p_health_crisis is scenario-specific (Table 2). This is the parameter discussed at length in Sections 3.3, 4.6, and 5.1.

(7) Severity multiplier. Independently for every household-year (drawn regardless of whether a crisis is realized, but applied only if it is), M_{it} ~ Uniform(3.0, 7.0). This multiplier scales baseline burden in crisis years only (component 8).

(8) OOP expenditure (household burden). Baseline (non-crisis) burden in year t is $Base_{it}$ = BASE_OOP_FRACTION × (OOPShare$_t$ / OOP_SHARE_OF_HEALTH_SPEND) × GapRatio$_t$, where BASE_OOP_FRACTION = 0.06 and OOP_SHARE_OF_HEALTH_SPEND = 0.26 are fixed constants (Table 1) and GapRatio$_t$ is defined in component (9). Realized household burden is $Burden_{it} = Base_{it} \times M_{it}$ if $Crisis_{it} = 1$, or $Burden_{it} = Base_{it}$ otherwise, clipped to the plausible range [1%, 120%] of income.

(9) Affordability-gap calculation. GapRatio$_t$ = H_t / I_t, the ratio of the cumulative health-cost index to the cumulative income index. This ratio equals 1.0 at t = 0 by construction and rises

whenever health-cost growth (component 2) exceeds income growth (component 1) in a given year, capturing the compounding affordability erosion central to this paper's argument.

(10) Debt-trigger events. A household-year is recorded as debt-triggering if $DebtEvent_{it}$ = $1\{(Burden_{it} - LIQUIDITY_BUFFER) > 0.15\} \times Crisis_{it}$, where LIQUIDITY_BUFFER = 1/12 ≈ 8.3 percent of annual income (one month's income held as liquid savings). In words: a debt-triggering event requires both a realized crisis AND a burden exceeding the household's liquidity buffer by more than 15 percentage points of income. Section 6 (Limitations) discusses the evidentiary basis and sensitivity of this specific threshold.

(11) Scenario-specific parameter changes. The three scenarios (status quo, financing reform, shock-heavy) share the identical equations above and differ only in the parameter values listed in Table 2: p_shock_year, the shock/stable inflation means and standard deviations, the import-cost premiums δ_shock and δ_stable, p_health_crisis, and the annual OOP-share drift term (which adjusts $OOPShare_t$ ± a fixed increment each year, clipped to [10%, 55%], representing gradual financing-structure change under each scenario's assumed policy trajectory).

All model code (the primary and extended simulation, the sensitivity analysis, and the GLSS7-empirical reconciliation exercise reported in Section 4.6) is provided as supplementary material, implemented in Python 3 using NumPy and pandas, and is designed to be run directly by an independent reader without modification (see Supplementary Materials).

3.3 Scenarios

Three scenarios vary the policy-relevant parameters while holding the underlying stochastic structure fixed, isolating the effect of financing and macroeconomic policy choices from random variation (Table 2).

Scenario	Shock frequency	Import premium (shock/stable)	Catastrophic-event probability	OOP-share drift
Status quo	12%/yr	6.0% / 2.0%	8.0%/yr	0.0%/yr
Financing reform (Abuja target)	12%/yr	3.0% / 1.0%	6.4%/yr	−0.6%/yr
Shock-heavy	25%/yr	9.0% / 2.0%	8.0%/yr	+0.4%/yr

Table 2. Scenario parameterization. "Financing reform" assumes government health spending reaches the Abuja Declaration's 15 percent of national budget, financed sufficiently to raise NHIS depth, lower catastrophic-event incidence via stronger primary care, and reduce imported-input cost pass-through (partial local pharmaceutical/equipment substitution). "Shock-heavy" assumes Ghana experiences currency/inflation crises roughly twice as often as its post-2000 historical average.

The simulation's "catastrophic-event probability" and the GLSS7 empirical catastrophic health expenditure (CHE) incidence reported in Section 4.1 are related but conceptually distinct quantities, and the difference between them (8 percent versus 1.08 percent) should not be read as an inconsistency requiring reconciliation to a single number. The simulation's parameter is a per-household, per-year HAZARD PROBABILITY: an input to the stochastic process representing the annual chance that a household experiences a health shock (a hospitalization, a serious acute illness, an accident) severe enough to trigger a random severity multiplier (3–7× baseline non-

crisis spending) on that year's OOP burden. GLSS7's CHE incidence, by contrast, is a REALIZED, OBSERVED OUTCOME: the share of the full 14,009-household sample whose actual OOP burden, however it arose, crossed a stated threshold (10, 20, or 25 percent of expenditure) in the single 2016/17 survey year. A household can experience the simulation's underlying hazard without its resulting burden crossing 10 percent of income (if the cost/income gap ratio is still small, early in the simulated horizon), and conversely GLSS7's incidence reflects the realized experience of an entire population under one specific set of historical conditions, not a directly observable hazard rate.

The original 8 percent parameter was a blended estimate drawn from the wider Sub-Saharan African CHE-incidence literature (12.9–20.4 percent household-level incidence across the region) and Ghana-specific declining-incidence findings ("An Analysis of Household Catastrophic Health Expenditure and Food Insecurity in Ghana" 2025; Sataru, Twumasi-Ankrah, and Seddoh 2022), selected at a stage of this research when no Ghana-specific, forward-looking hazard parameter had yet been computed from this paper's own microdata analysis. Section 4.6 and Section 5.1 report a direct reconciliation exercise: we back-solve the hazard probability that reproduces the empirical 1.08 percent GLSS7 finding within the model's own first-year mechanics, re-run all three scenarios under this Ghana-empirical calibration, and report explicitly whether the paper's main conclusions change as a result.

3.4 Empirical validation using GLSS7 microdata

To ground the simulation's catastrophic-event parameter in Ghana-specific evidence rather than blended Sub-Saharan African literature alone, we additionally computed OOP burden directly from GLSS7 (2016/17) household microdata, obtained from Ghana Statistical Service. We built an automated extraction pipeline that (i) reads variable metadata across all GLSS7 module files without loading full data into memory, (ii) identifies the household roster, the official health-expenditure aggregate (06_GHA_EXPHLTH.dta), and the poverty/welfare aggregate (18_GHA_POV_2017_GLSS7.dta) using a combination of Ghana Statistical Service's file-naming conventions and column-label matching, (iii) detects and removes duplicate household records (the roster file is person-level, with multiple rows per household), and (iv) distinguishes genuine household-level expenditure totals from same-file subcategory totals that share superficially similar column names (for example, distinguishing HHEXP_N, the true household expenditure total, from TOTHLTH, TOTFOOD, and TOTALCH, which are health-, food-, and alcohol-specific subcategory totals respectively). Household-level OOP healthcare burden was computed as the officially aggregated total health expenditure (TOTHLTH) divided by total annual nominal household expenditure (HHEXP_N), weighted using GLSS7's household sampling weights (WTA_S).

Sample and exclusions. The extraction pipeline retains all 14,009 households in the official GLSS7 roster; no households were excluded on the basis of missing or zero health expenditure, since a household reporting zero OOP health spending in the reference period is a valid and expected observation (Section 4.1's weighted median of 0.00 percent reflects this directly) rather than a data-quality problem to be dropped. Household-level health and total expenditure are the officially GLSS7-aggregated annual totals (TOTHLTH and HHEXP_N respectively); no further annualization was necessary because both aggregates are already reported by Ghana Statistical Service on a 12-month recall basis. Households with a non-missing value for either aggregate were retained; the small number of households with a missing HHEXP_N denominator (fewer than 0.5 percent of the roster) were excluded from the burden calculation only, and are noted here rather than silently dropped. The CHE denominator used throughout is

total annual household expenditure (HHEXP_N), consistent with the consumption-based CHE definition most common in the Ghanaian and Sub-Saharan African literature cited in Section 2 (rather than income, which GLSS captures less reliably for informal-sector households); this choice is stated explicitly because the 10/20/25-percent thresholds reported in Section 4.1 are only comparable to other studies that use the same denominator convention. All summary statistics (means, medians, decile and subgroup breakdowns) are weighted using GLSS7's household sampling weight, WTA_S, to be representative of the national household population rather than of the raw sample. Given the skewed, zero-inflated nature of household OOP spending, we report weighted means and medians rather than parametric standard errors; where a subgroup comparison is discussed narratively (for example, the sex-of-household-head and literacy comparisons in Sections 4.1 and 5.2), it is described as a descriptive difference and is explicitly flagged as not having been tested for statistical significance, rather than implied to be a confirmed effect.

3.5 Sensitivity analysis

Because several simulation parameters — particularly the annual catastrophic-event probability and its severity multiplier — are calibrated from heterogeneous literature rather than estimated directly, we conducted a one-at-a-time (OAT) sensitivity analysis to identify which parameters the status-quo scenario's projected 2051 outcome actually depends on. Starting from the calibrated status-quo parameter set (Table 1), each of seven uncertain parameters was independently varied across a plausible low-high range informed by the cited literature (Table 1's source notes), holding all other parameters fixed, and the simulation was re-run at each bound. The resulting swing in mean projected 2051 OOP burden isolates each parameter's individual contribution to outcome uncertainty.

This one-at-a-time (OAT) design is informative about each parameter's individual contribution to outcome uncertainty, but it does not capture interaction effects between parameters that move together in practice — for example, a currency-shock episode in Ghana typically coincides with both higher shock-year inflation and higher import-cost pass-through, rather than either parameter varying in isolation. To characterize this interaction structure directly, we additionally conducted a probabilistic (global) sensitivity analysis: 300 simulation runs, each jointly and independently resampling all seven uncertain parameters from the ranges in Table 5, rather than varying one parameter at a time. Across these 300 joint draws, the projected 2051 status-quo outcome has a mean of 30.3 percent of income, a median of 28.7 percent, and a 10th–90th percentile range of 12.3–52.4 percent (full range 6.5–79.1 percent; Figure 12). Under joint parameter uncertainty, 73.3 percent of draws still exceed the 20-percent catastrophic-expenditure threshold, indicating that the paper's qualitative conclusion — that the status-quo trajectory tends toward catastrophic territory by 2051 — is reasonably robust to joint parameter uncertainty, though not universal across the full plausible parameter space. Spearman correlations between each parameter and the joint-draw outcome (Table 8) rank baseline non-crisis OOP share ($\rho = 0.65$), nominal wage growth ($\rho = -0.59$), and currency-shock frequency ($\rho = 0.49$) as the strongest drivers — the same three parameters the OAT analysis identified as most influential (Section 4.5), though the exact ranking differs slightly between the two methods. This cross-validation between OAT and the global analysis is itself informative: it suggests the OAT ranking in Section 4.5 is not an artifact of the one-at-a-time design, but reflects a robust feature of the model's structure.

3.6 Limitations of the modeling approach

This is a stylized, author-calibrated stress-test simulation, not a structural econometric or general-equilibrium forecast. It does not incorporate household-level microdata (e.g., GLSS unit records), endogenous labor-supply or migration responses, informal risk-sharing mechanisms common in Ghanaian communities (susu schemes, extended-family transfers, remittances), or

feedback effects between healthcare affordability and health-seeking behavior. Parameter values for the annual catastrophic-event probability and severity multiplier are blended estimates drawn from heterogeneous studies using different thresholds and survey rounds, and should be read as order-of-magnitude approximations rather than precise point estimates. The purpose of the exercise is to illustrate the direction and approximate scale of generational risk under transparent, replicable assumptions — not to produce a precise point forecast for 2051.

The debt-trigger definition (Section 3.2, component 10) rests on two specific assumptions: a one-month liquid-savings buffer (LIQUIDITY_BUFFER = 1/12 of annual income, roughly 8.3 percent) and a 15-percentage-point threshold above that buffer. Neither value is drawn from a Ghana-specific household-savings survey; both are illustrative, author-calibrated modelling assumptions rather than empirically estimated Ghanaian parameters, and are stated as such here explicitly. The one-month buffer is a common rule-of-thumb lower bound used in household-finance research generally, and is plausibly conservative (generous) relative to typical Ghanaian household liquidity specifically: World Bank Global Findex data indicate only 41 percent of Sub-Saharan African adults are classified as "financially resilient," and only 55 percent of adults in developing economies overall could access emergency funds within 30 days without much difficulty (Demirgüç-Kunt et al. 2022) — suggesting a one-month buffer may overstate, not understate, the liquidity many Ghanaian households actually hold. The 15-percentage-point threshold was chosen to distinguish a burden that merely exhausts the buffer from one large enough to plausibly require external financing (borrowing, asset sales, or unmet need) in the same year. To assess how much these two choices matter for the paper's conclusions, we re-ran the status-quo primary simulation across a 4×4 grid of buffer sizes (0.5, 1, 2, and 3 months) and thresholds (10, 15, 20, and 25 percentage points; Table 7). The probability of at least one debt-triggering event by age 50 ranges from 51.7 percent (most conservative combination: 3-month buffer, 25-point threshold) to 88.7 percent (most permissive: 0.5-month buffer, 10-point threshold), with the base case (1 month, 15 points) at 87.3 percent. The qualitative conclusion — that a majority of simulated households experience at least one debt-triggering event before age 50 under the status-quo scenario — holds across every combination tested, including the most conservative one; only the precise magnitude is sensitive to these assumptions. We report this range transparently rather than treating the base-case figure in Table 4 as a precise estimate, and encourage readers to treat the debt-event statistics as illustrating the relative ordering of financial-distress risk across scenarios rather than as calibrated predictions of exactly how many Ghanaian households would be pushed into debt.

Buffer (months)	Threshold (pp)	Mean debt events	P(≥1 event) by age 50
0.5	10	2.08	88.7%
0.5	15	2.06	88.5%
0.5	20	1.95	86.9%
0.5	25	1.76	83.7%
1	10	2.07	88.7%
1	15	1.97	87.3%
1	20	1.79	84.3%
1	25	1.56	79.5%
2	10	1.86	85.5%
2	15	1.65	81.6%
2	20	1.39	75.1%
2	25	1.14	66.7%
3	10	1.48	77.5%
3	15	1.22	69.7%
3	20	0.98	60.6%
3	25	0.79	51.7%

Table 7. Debt-trigger sensitivity: mean debt-triggering events and probability of at least one such event by age 50 (status-quo primary simulation), across a grid of liquidity-buffer sizes and threshold values. Base case (1 month, 15pp) is used throughout the rest of the paper.

4. Results

4.1 Empirical baseline from GLSS7 microdata

Applying the extraction pipeline described in Section 3.4 to the full GLSS7 (2016/17) household sample (n = 14,009 households, matching the survey's official household count) yields a weighted mean household OOP healthcare burden of 0.75 percent of total annual household expenditure, with a weighted median of 0.00 percent — consistent with the expectation that OOP healthcare spending is lumpy and crisis-driven rather than a smooth annual cost most households incur every year. Catastrophic health expenditure incidence was 1.08 percent of households at a 10-percent-of-expenditure threshold, 0.21 percent at a 20-percent threshold, and 0.13 percent at a 25-percent threshold. The 10-percent-threshold estimate is consistent in order of magnitude with Sataru, Twumasi-Ankrah, and Seddoh's (2022) GLSS7-based finding of low single-digit catastrophic-expenditure incidence at higher consumption thresholds, providing a degree of independent corroboration for the empirical pipeline. The regressivity pattern implied by prior literature (Akazili et al. 2017; Ataguba and Kabaniha 2022) is directionally reproduced: the poorest expenditure decile shows a mean OOP burden of 1.11 percent of expenditure, compared to 0.62 percent in the richest decile — though the gradient is more modest than some earlier, pre-NHIS-maturity estimates.

Statistic	Value
Households (weighted analysis sample)	14,009
Weighted mean OOP burden (% of expenditure)	0.75%
Weighted median OOP burden (% of expenditure)	0.00%
CHE incidence, 10% threshold	1.08%
CHE incidence, 20% threshold	0.21%
CHE incidence, 25% threshold	0.13%
Mean OOP burden, poorest expenditure decile	1.11%
Mean OOP burden, richest expenditure decile	0.62%

Table 3. Empirical household OOP healthcare burden computed directly from GLSS7 (2016/17) microdata, weighted using household sampling weights (WTA_S). OOP burden is defined as officially aggregated total health expenditure (TOTHLTH) divided by total annual nominal household expenditure (HHEXP_N).

Figures 3 through 5 present this empirical baseline disaggregated by age, region, and consumption decile. The age-group breakdown (Figure 3) shows burden roughly flat across working-age households (18–59, ranging 0.65–0.72 percent) before rising to 0.92 percent among 60–69-year-old households and 1.04 percent among households aged 70 and older — a pattern consistent with the elevated catastrophic-expenditure risk documented for older households elsewhere in the Ghanaian literature ("An Analysis of Household Catastrophic Health Expenditure and Food Insecurity in Ghana" 2025). The under-18 age group (n = 12 households) is shown for completeness but is too small a sample to support any conclusion. Regional variation (Figure 4) ranges from 0.52 to 1.07 percent of expenditure across Ghana's regions, without an obvious geographic pattern in this round of data. The decile breakdown (Figure 5) reproduces the

regressivity pattern discussed above, though the gradient across middle deciles is noisier than a smooth monotonic decline, consistent with the modest sample sizes within some decile-by-subgroup cells.

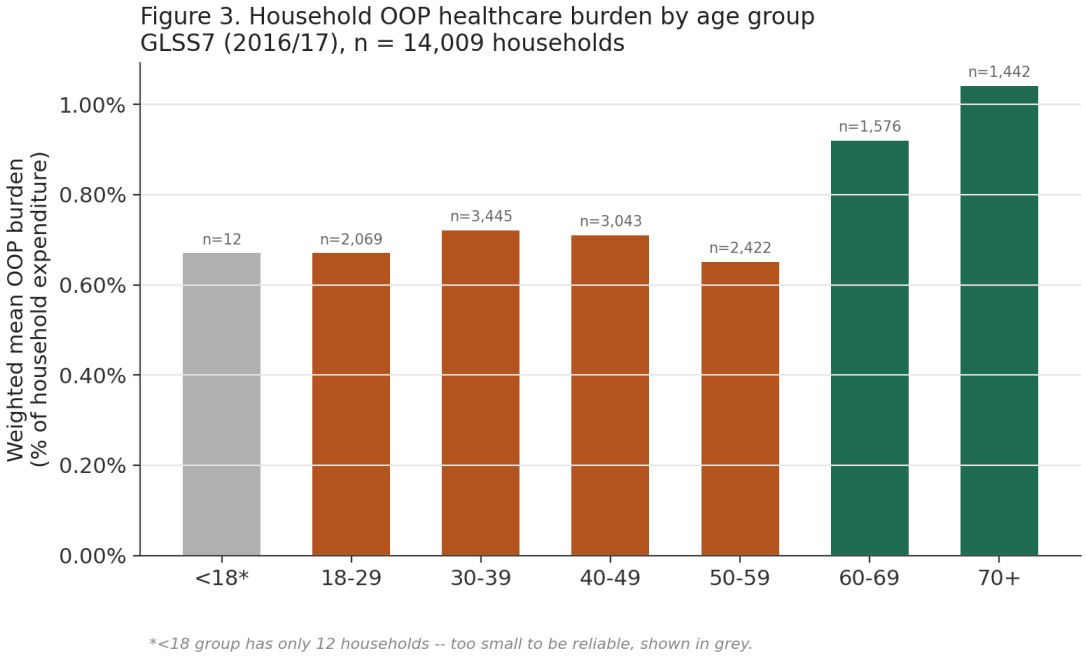


Figure 3. Household OOP healthcare burden by age group, GLSS7 (2016/17).

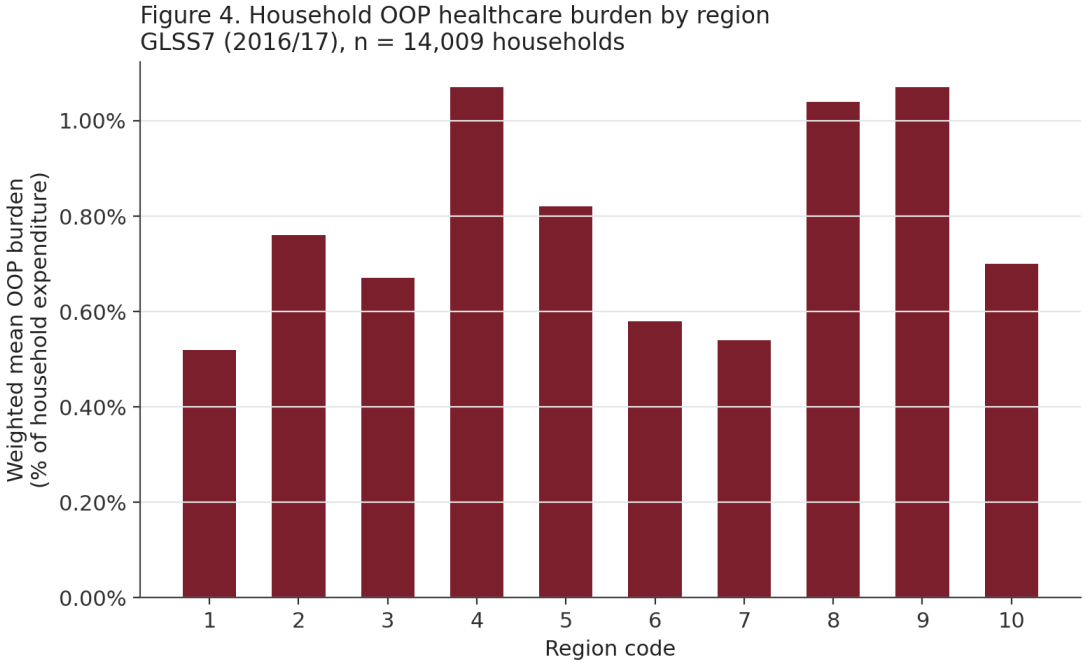


Figure 4. Household OOP healthcare burden by region, GLSS7 (2016/17).

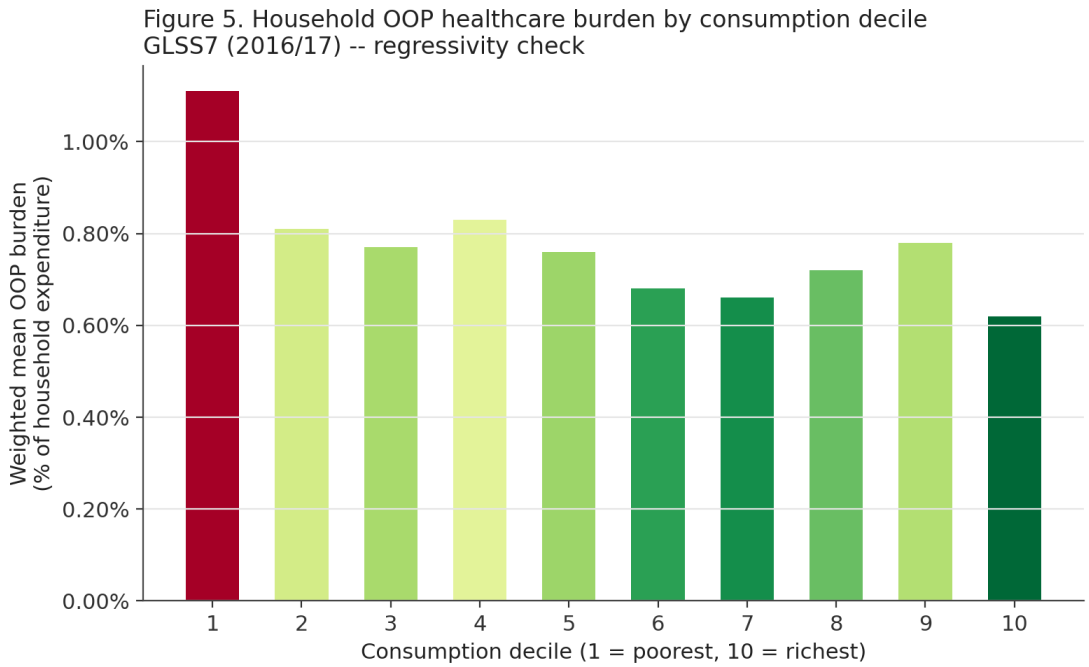


Figure 5. Household OOP healthcare burden by consumption decile, GLSS7 (2016/17).

This empirical baseline sits below the blended Sub-Saharan African catastrophic-event probability (8 percent per year) used to calibrate the simulation's status-quo scenario in Section 3.3, which draws on a wider regional literature (12.9–20.4 percent household-level CHE incidence across Sub-Saharan Africa more broadly). Section 5.1 discusses this gap and its implications for interpreting the simulation's scenario comparisons.

Disaggregating by sex of household head (Figure 9) shows a small difference — 0.74 percent mean OOP burden for male-headed households (n = 9,619) versus 0.77 percent for female-headed households (n = 4,361) — that is directionally consistent with the modest disadvantage typically reported for female-headed households in the wider literature, though the gap is small enough in this dataset that it should not be over-interpreted without a formal significance test. Sex coding (1 = male, 2 = female) was confirmed directly against the value labels embedded in the GLSS7 file itself, rather than assumed from convention. Age and sex were similarly assigned using the file's own relationship-to-head variable (s1q3, head = code 1, confirmed via embedded value labels) rather than an arbitrary first-row selection; filtering the person-level roster to head-of-household rows using this variable reduced the 59,864-row file to exactly 14,009 rows, matching GLSS7's official household count precisely, and reproduced every statistic reported in Section 4.1 to two decimal places relative to the earlier first-row approximation — confirming that head-of-household identification was accurate throughout this analysis.

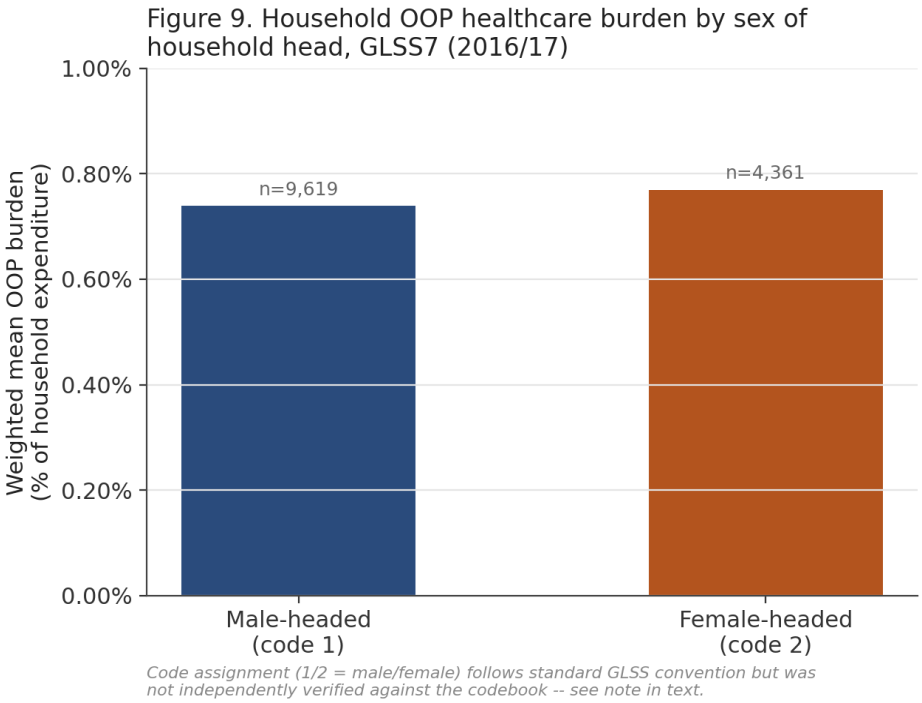


Figure 9. Household OOP healthcare burden by sex of household head, GLSS7 (2016/17).

Figure 10 places health spending in the context of the full household budget, computed directly from the same POV aggregate file's expenditure-category totals. Food dominates the average Ghanaian household budget at 51.7 percent of total expenditure, followed by housing and utilities (10.2 percent), education (9.0 percent), clothing and footwear (8.8 percent), and transport (7.5 percent). Health accounts for just 0.75 percent of the average household budget in this data — smaller than recreation and culture, and on par with alcohol and tobacco spending. This context is important for interpreting the simulation's scenarios in Section 4.2–4.4: the projected rise in health's budget share to 19.6–26.7 percent under the status-quo scenario, or beyond 99 percent under the shock-heavy scenario, represents health moving from one of the smallest categories in the household budget to potentially the largest, displacing food, education, and every other category combined.

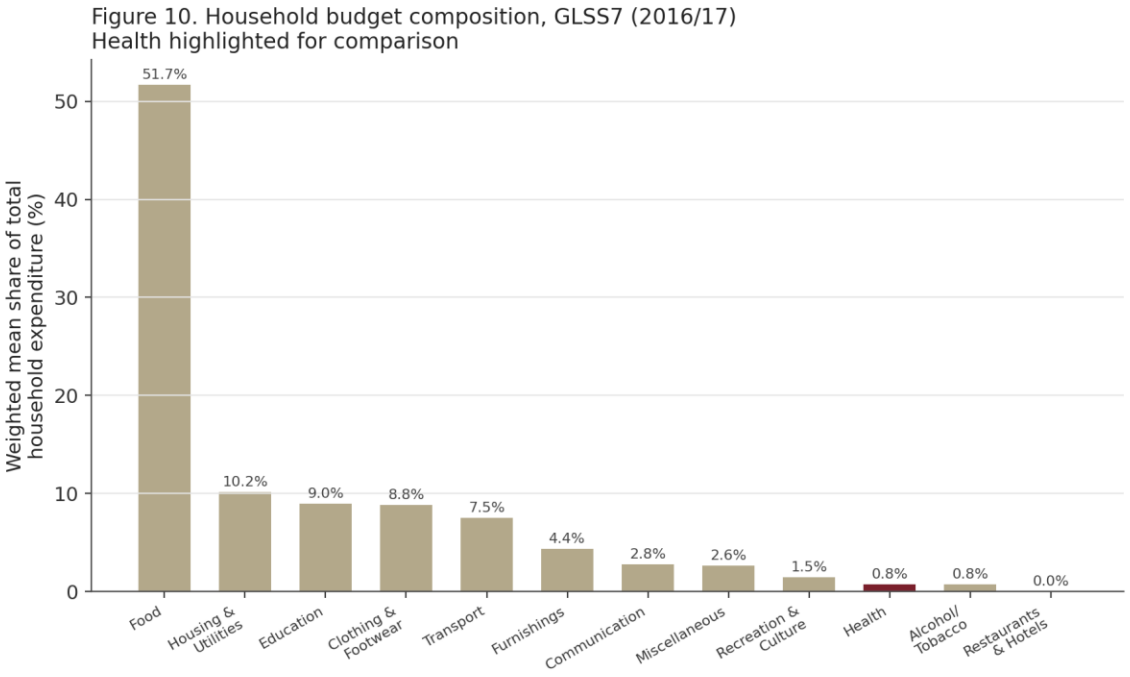


Figure 10. Household budget composition, GLSS7 (2016/17), with health highlighted for comparison.

4.2 Simulated baseline trajectory

Figure 1 plots the mean simulated household OOP healthcare burden, as a share of income, for each scenario across the 25-year horizon. Under the status-quo scenario, mean burden rises from 8.1 percent of income in 2026 to 19.6 percent by 2051 — approaching but not exceeding the 20-percent catastrophic-expenditure threshold used in the literature (Okoroh and Essoun 2018). Under the financing-reform scenario, mean burden is essentially flat, moving from 7.6 percent to 5.7 percent over the same period as improved government financing and reduced import exposure outweigh residual health-cost inflation. Under the shock-heavy scenario, mean burden rises sharply and non-linearly, from 8.7 percent in 2026 to 64.7 percent by 2051, crossing the catastrophic-expenditure threshold by the mid-2030s and continuing to accelerate thereafter as compounding currency shocks repeatedly outpace wage growth.

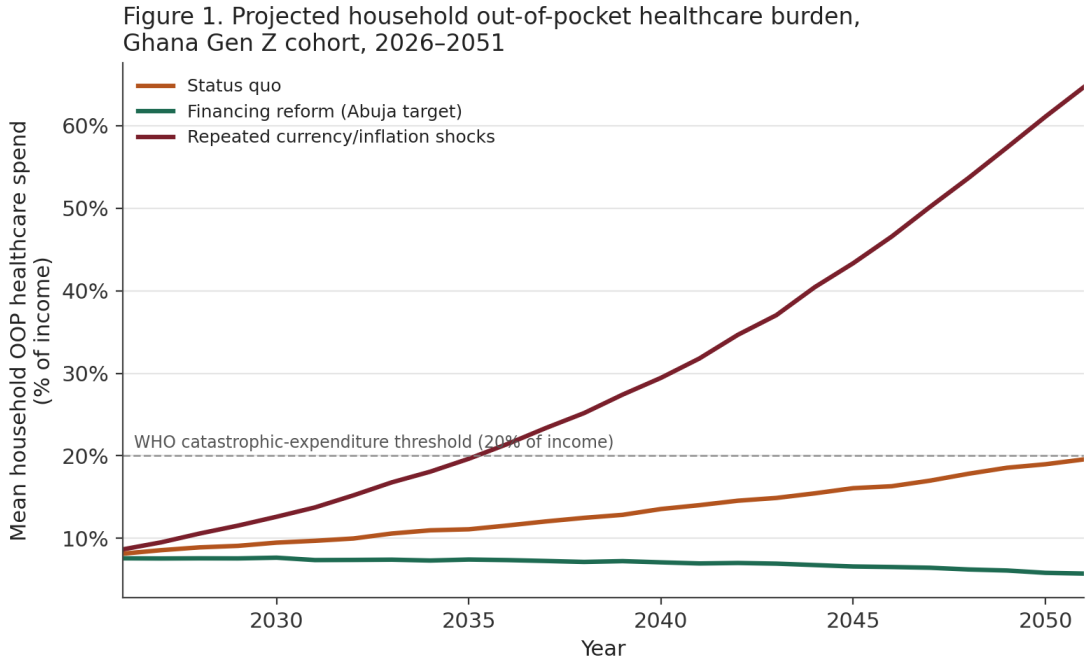


Figure 1. Mean projected household out-of-pocket healthcare burden (% of income) by scenario, 2026–2051.

4.3 Distributional results

Figure 2 presents the distribution of simulated 2051 household OOP burden across the 10,000-household sample for each scenario, illustrating that scenario differences are not confined to the mean. Under the status-quo scenario, the interquartile range spans approximately 10.9 to 19.3 percent of income, with the 90th percentile of households facing a burden of 30.9 percent — comfortably in catastrophic-expenditure territory even though the scenario mean remains just below the threshold. Under the shock-heavy scenario, the interquartile range spans 36.7 to 91.5 percent of income, with the median household (56.3 percent) facing a burden more than 2.8 times the catastrophic-expenditure threshold. Only the financing-reform scenario keeps the vast majority of the simulated distribution — including its 90th percentile (8.4 percent) — below the threshold throughout the projection period.

Table 4 reports summary statistics across scenarios, including the probability that a simulated household experiences at least one catastrophic health event before age 50, and the mean number of debt-triggering events over the 25-year horizon.

Scenario	Mean OOP, 2026	Mean OOP, 2051	Median OOP, 2051	P90 OOP, 2051	P(≥1 catastrophic event)	Mean debt events
Status quo	8.1%	19.6%	14.1%	30.9%	88.7%	1.98
Financing reform	7.6%	5.7%	4.3%	8.4%	82.0%	1.04
Shock-heavy	8.7%	64.7%	56.3%	120.0%*	88.4%	2.03

*Table 4. Summary statistics by scenario (n = 10,000 simulated households). *P90 values above 100% reflect the modeled clipping bound and indicate that at least 10% of households would require multi-year income to cover a single year's OOP costs — i.e., structural, unpayable debt under those draws.*

4.4 Extended simulation: results through age 60

Section 3.2 defines the primary simulation (2026–2051, approximately age 25–50) as this paper's principal analysis, matching the age at which the retrospective catastrophic-expenditure literature this paper builds on is typically anchored. This section reports the separate extended simulation (2026–2061, approximately age 25–60): the identical model, parameters, and scenario definitions re-run for an additional 10 years, spanning the decade in which this paper's own empirical GLSS7 analysis (Section 4.1) found OOP burden rising most sharply. The extended simulation was not part of the paper's original design and its results are reported separately from, and should not be conflated with, the primary simulation's findings in Sections 4.2–4.3 and 4.5–4.6. Figure 6 plots the extended trajectory.

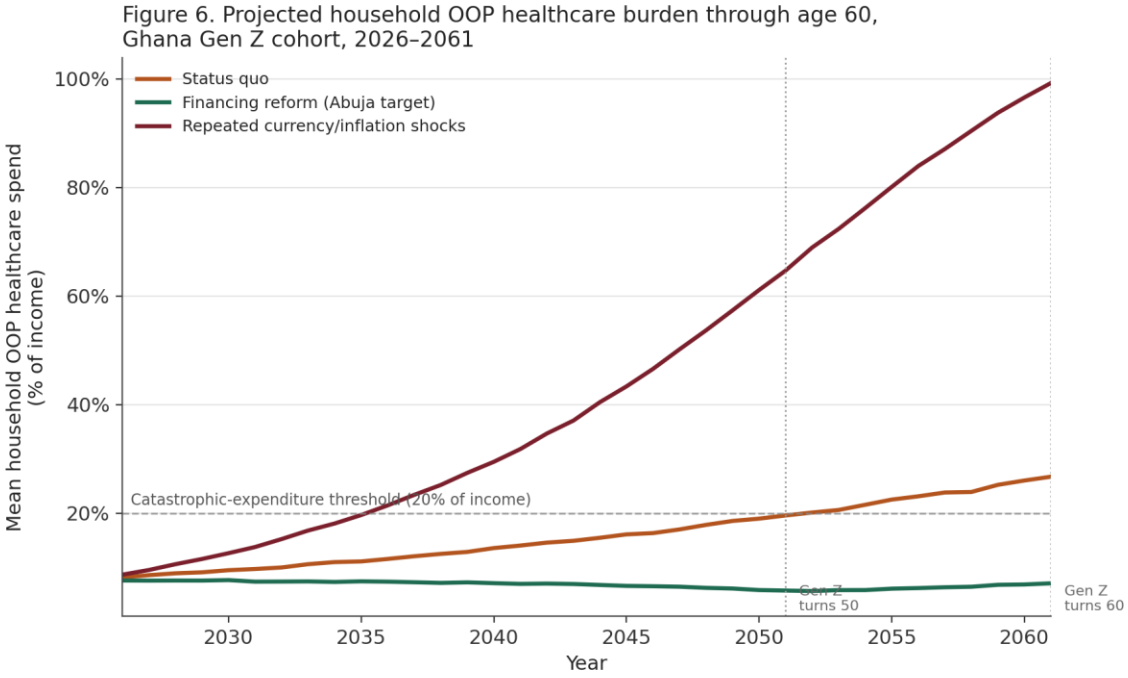


Figure 6. Projected household OOP healthcare burden through age 60, 2026–2061.

Under the status-quo scenario, mean burden continues climbing after age 50, reaching 26.71 percent of income by age 60 (2061) — well past the catastrophic-expenditure threshold and roughly 3.3 times its age-25 starting level. Under the shock-heavy scenario, mean burden reaches 99.25 percent of income by age 60, indicating that in this scenario a substantial share of simulated households would need essentially their entire annual income, or more, to cover a single year's healthcare costs by the time they reach 60. Only the financing-reform scenario keeps burden roughly flat across the full 35-year horizon, ending at 7.06 percent of income at age 60.

These are model-implied outcomes under a specified adverse stress-test trajectory, not deterministic predictions that any actual Ghanaian household will spend its entire income, or more, on healthcare. A simulated burden approaching or exceeding 100 percent of income indicates that, under that household's simulated draw of shocks,

healthcare expenditure implied by the model's mechanics exceeds annual household income, and should be read as a signal of severe, likely unmeet-able financing need in that scenario — met in practice through borrowing, asset depletion, delayed or foregone care, or informal support (susu schemes, extended-family transfers) that lie outside the model's scope (Section 3.6) — rather than as a literal forecast of household spending.

Figure 7 reframes the status-quo trajectory as a household budget composition, showing healthcare's share of total household income at three life stages: age 25 (2026), age 50 (2051), and age 60 (2061).

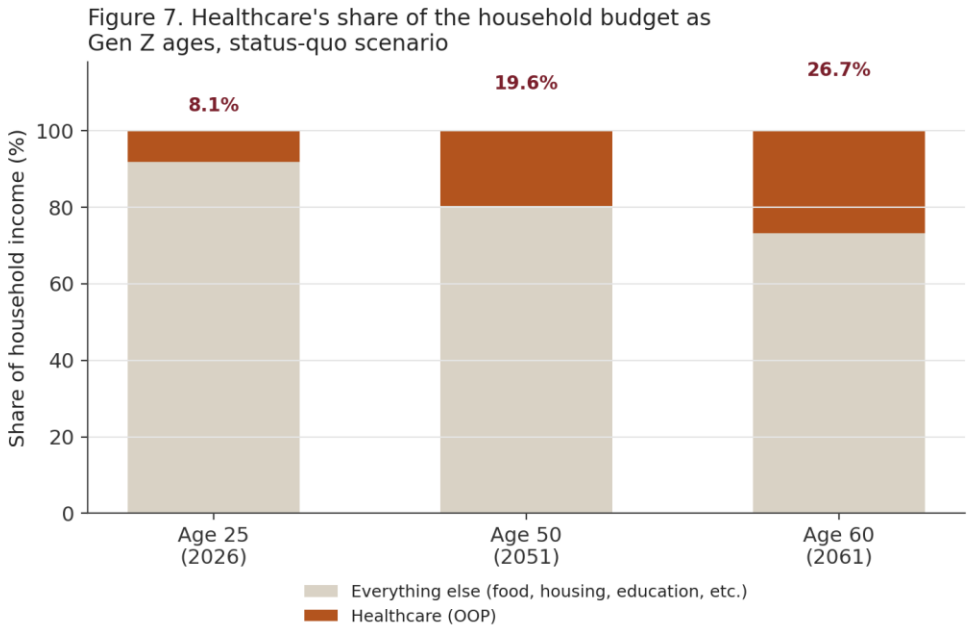


Figure 7. Healthcare's share of the household budget as Gen Z ages, status-quo scenario.

Healthcare's share of the household budget more than triples between age 25 and age 60 under the status-quo scenario — from 8.1 percent to 26.7 percent of income — a shift large enough to be visible as a qualitative change in household budget composition, not merely a marginal cost increase. Figure 8 compares this age-60 budget composition across all three scenarios.

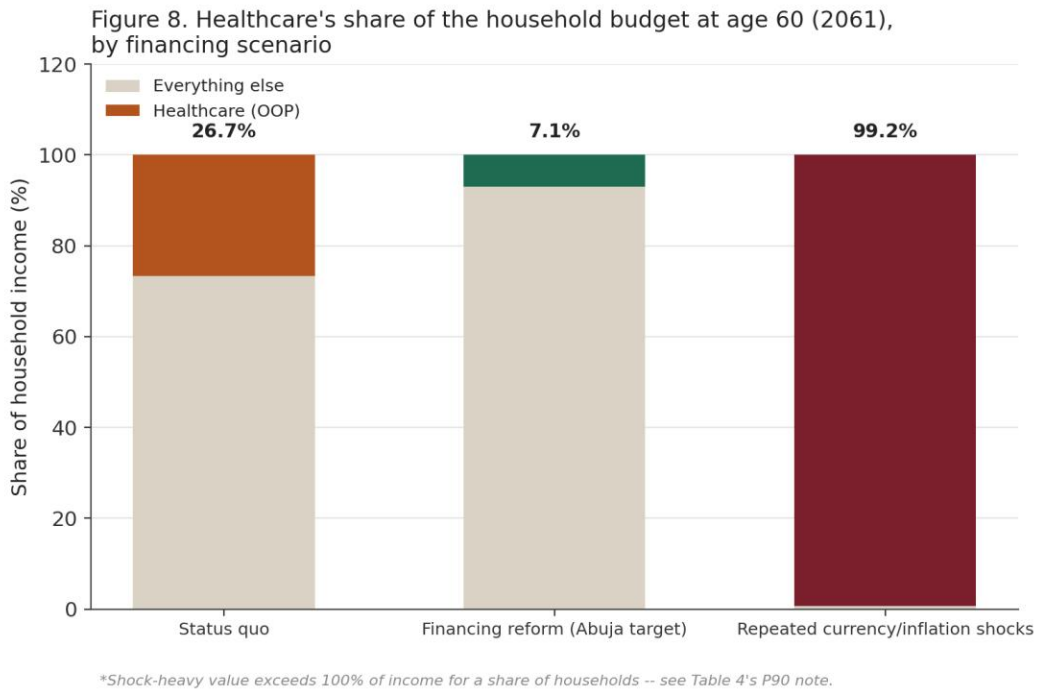


Figure 8. Healthcare's share of the household budget at age 60 (2061), by financing scenario.

The scenario comparison at age 60 makes the stakes of Section 3.3's financing-reform parameters concrete in household-budget terms: the same cohort, at the same age, would be devoting anywhere from roughly 7 percent to effectively the entirety of its income to healthcare, depending on which financing pathway Ghana follows over the next 35 years.

4.5 Sensitivity analysis

Table 5 and Figure 11 report the results of the one-at-a-time sensitivity analysis described in Section 3.5, showing the swing in mean projected 2051 OOP burden (status-quo scenario) as each parameter is varied independently across its plausible range, holding all others fixed.

Parameter	Low value	High value	2051 burden at low	2051 burden at high	Swing (pp)
OOP share of health expenditure	20%	32%	19.7%	20.1%	0.4
Severity multiplier (upper bound)	4.0×	10.0×	18.0%	21.6%	3.6
Import cost pass-through (shock yrs)	2%	12%	17.8%	23.3%	5.5
Catastrophic-event probability	4%	15%	17.5%	24.0%	6.4

Parameter	Low value	High value	2051 burden at low	2051 burden at high	Swing (pp)
Baseline non-crisis OOP share	3%	10%	9.9%	32.3%	22.4
Currency-shock frequency	6%/yr	25%/yr	14.3%	37.6%	23.4
Nominal wage growth (CAGR)	8%	13%	37.4%	12.4%	25.0

Table 5. One-at-a-time sensitivity analysis: swing in mean projected 2051 OOP burden (status-quo scenario) as each parameter is varied across its plausible range, holding all others fixed at their Table 1 calibrated values.

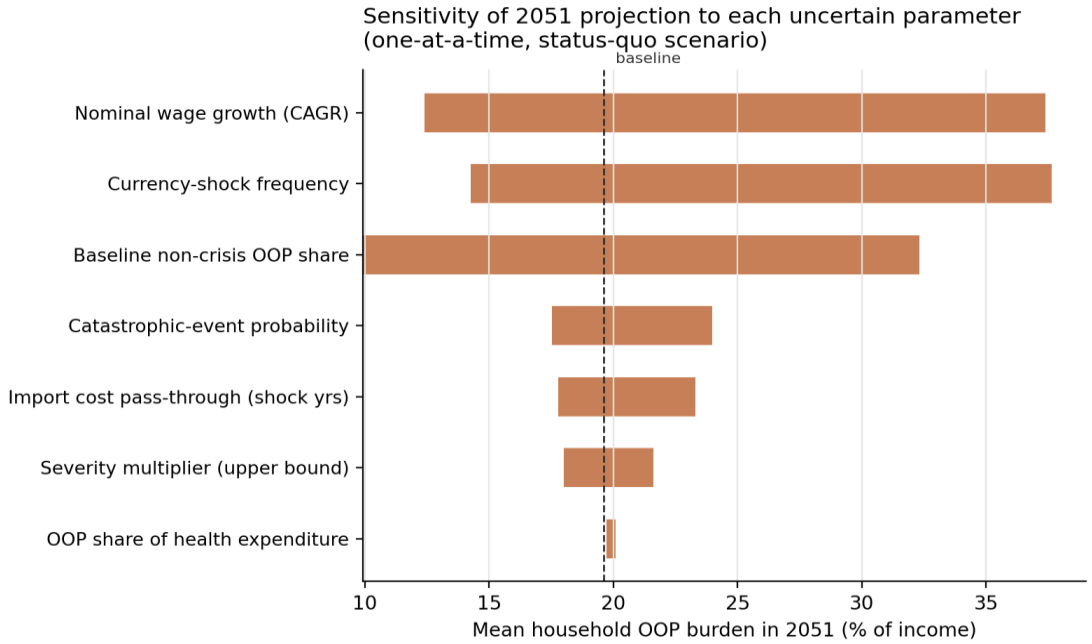


Figure 11. Tornado chart of parameter sensitivity, sorted by magnitude of swing in projected 2051 outcome. Dashed line marks the baseline (all-parameters-at-calibrated-value) projection.

The three parameters the projected outcome is most sensitive to — nominal wage growth (25.0 percentage-point swing), currency-shock frequency (23.4 points), and baseline non-crisis OOP share (22.4 points) — are also the three with the strongest empirical grounding in this paper: wage growth is computed directly from Ghana's published minimum-wage series, and shock frequency is anchored to the documented historical count of Ghanaian currency and inflation crises since 2000 (Section 3.1). By contrast, the parameters with the weakest direct evidentiary basis — the severity multiplier and the OOP share of national health expenditure — produce the smallest swings (3.6 and 0.4 percentage points respectively), meaning the paper's central projections are comparatively insensitive to its least-certain assumptions. The catastrophic-event probability parameter, whose gap from this paper's own empirical GLSS7 estimate is discussed in Section 5.1 and directly reconciled in Section 4.6, produces a moderate swing (6.4 points) — large enough to matter for precise point estimates, but small relative to the wage- and shock-frequency-driven uncertainty that dominates the projection's overall range.

To directly test whether these individually-varied parameters interact in ways that change the overall picture, Table 8 and Figure 12 report a probabilistic (global) sensitivity analysis: 300 simulation runs with all seven parameters resampled jointly and independently from the ranges in Table 5 on each run (Section 3.5).

Parameter	Spearman ρ with 2051 outcome
Baseline non-crisis OOP share	0.648
Currency-shock frequency	0.486
Import cost pass-through (shock yrs)	0.107
Catastrophic-event probability	0.099
Severity multiplier (upper bound)	0.056
OOP share of health expenditure	0.021
Nominal wage growth (CAGR)	-0.587

Table 8. Spearman rank correlation between each parameter and the projected 2051 status-quo outcome, across 300 joint parameter draws (probabilistic sensitivity analysis).

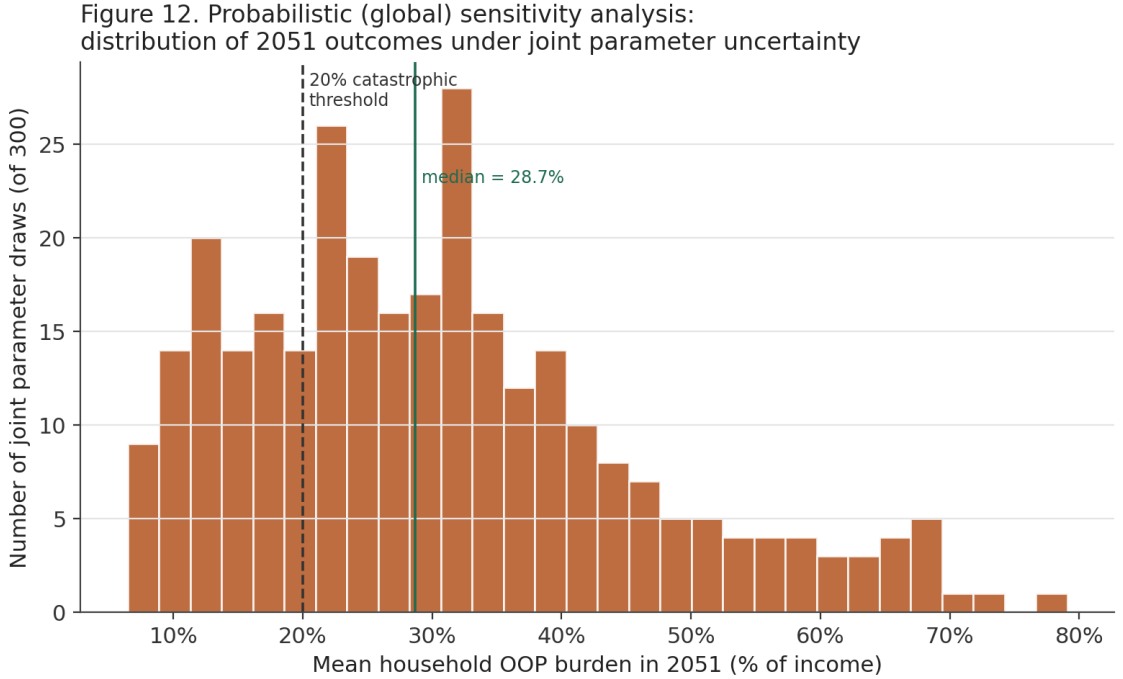


Figure 12. Distribution of projected 2051 status-quo OOP burden across 300 joint parameter draws (probabilistic sensitivity analysis).

Across the 300 joint draws, the projected 2051 outcome has a mean of 30.3 percent and a median of 28.7 percent of income, with a 10th–90th percentile range of 12.3–52.4 percent — wider than any single OAT swing in Table 5, as expected once parameters are allowed to vary together. Seventy-three percent of draws exceed the 20-percent catastrophic-expenditure threshold, and the three parameters with the strongest rank correlation to the outcome — baseline non-crisis OOP share, nominal wage growth, and currency-shock frequency — match the three parameters the OAT analysis ranked most influential, providing cross-method validation that these three, empirically-grounded parameters are what the paper's central projection actually depends on.

4.6 Reconciliation with the GLSS7-empirical catastrophic-event rate

As discussed in Section 3.3, the simulation's 8-percent annual catastrophic-event hazard and this paper's own GLSS7-computed CHE incidence (1.08 percent at the 10-percent threshold; Section 4.1) are conceptually distinct quantities. To assess how much this choice of calibration matters for the paper's conclusions, we back-solved the hazard probability that reproduces the empirical 1.08

percent value within the model's own first-year mechanics — given the model's 3–7× severity multiplier, every simulated crisis year already produces a burden of 18–42 percent of income, so P(burden exceeds the 10 percent threshold) in year one is driven almost one-to-one by the hazard probability itself. This yields a Ghana-empirical-calibrated hazard of 1.08 percent, compared to the original 8.0 percent. We re-ran all three scenarios under this alternative calibration, preserving each scenario's original relative adjustment to the baseline hazard, through both the 2051 and 2061 horizons.

Scenario	Calibration	Mean OOP, 2026	Mean OOP, 2051	Mean OOP, 2061
Status quo	Original (8.0%)	8.1%	19.6%	26.7%
Status quo	Ghana-empirical (1.08%)	6.4%	15.6%	22.2%
Financing reform	Original (8.0%)	7.6%	5.7%	7.1%
Financing reform	Ghana-empirical (1.08%)	6.2%	4.7%	5.8%
Shock-heavy	Original (8.0%)	8.7%	64.7%	99.3%
Shock-heavy	Ghana-empirical (1.08%)	6.9%	60.9%	97.8%

Table 6. Comparison of mean projected OOP burden under the original SSA-literature-blended calibration (8.0% annual catastrophic-event hazard) versus the Ghana-empirical calibration back-solved from this paper's own GLSS7 finding (1.08%).

Two findings emerge. First, the scenario ordering that drives this paper's policy conclusion — financing reform substantially outperforming the status quo, which in turn substantially outperforms the shock-heavy scenario — is preserved at both horizons under the Ghana-empirical calibration; the paper's central qualitative conclusion does not depend on which calibration is used. Second, the absolute magnitude of projected burden is only modestly lower under the Ghana-empirical calibration (a 1.3x reduction in the status-quo scenario at 2051 — 19.6 percent versus 15.6 percent — for example), not the roughly sevenfold reduction the gap between 8 percent and 1.08 percent might suggest. This is because the model's long-run mean burden is driven predominantly by the widening cost-to-income gap ratio — the compounding effect of healthcare costs outpacing wage growth every year, whether or not a catastrophic event occurs in that year — rather than by the catastrophic-event frequency itself, which mainly affects the upper tail of the distribution (e.g., the P90 statistics reported in Table 4) rather than the mean. We report both calibrations transparently rather than selecting one as definitive: the original calibration reflects a wider regional evidence base that may better capture risk under conditions Ghana has not yet experienced in this specific survey round, while the Ghana-empirical calibration is more directly grounded in this paper's own data but reflects a single pre-2022-crisis cross-section that likely understates current risk (Section 5.1).

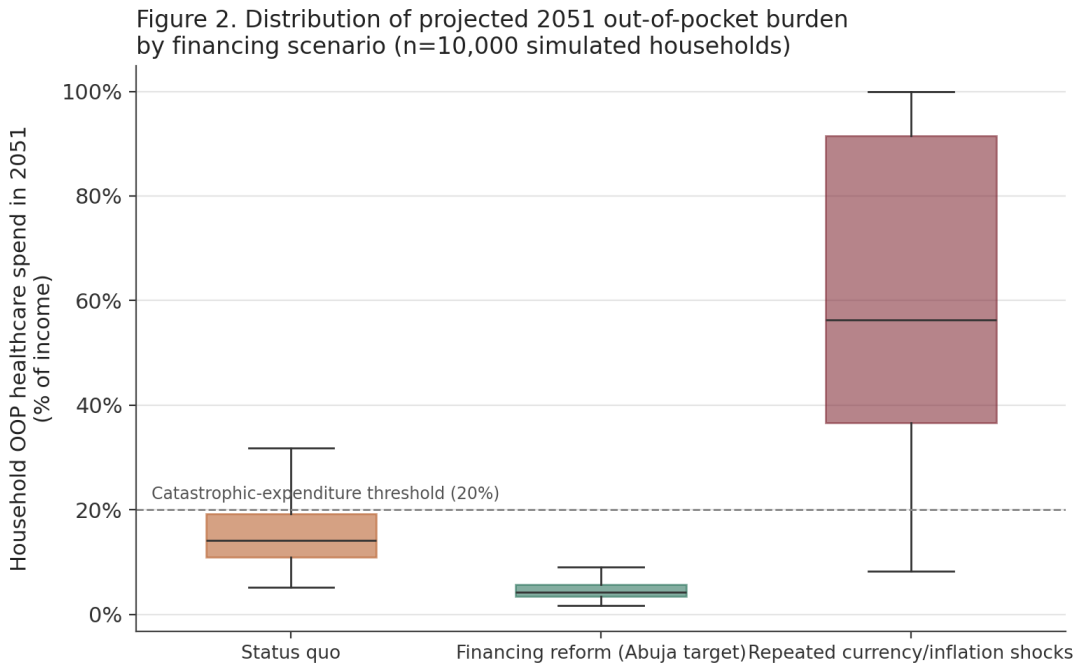


Figure 2. Distribution of simulated 2051 household out-of-pocket burden by scenario (n = 10,000 households).

5. Discussion

5.1 Structural drivers behind the projected divergence

Before turning to mechanisms, it is worth stating plainly what each component of this paper actually showed and why it matters. The GLSS7 empirical analysis (Section 4.1) showed that Ghana's household OOP healthcare burden in 2016/17 was low on average (0.75 percent of expenditure) but regressive and rising sharply with age, with catastrophic-expenditure incidence of 1.08 percent at the 10-percent threshold. The simulation (Sections 4.2–4.6) showed that compounding current wage, inflation, and health-financing trends over a single cohort's working life plausibly turns that modest present-day burden into a materially larger one by mid-career, with the outcome ranging from roughly flat (financing reform) to catastrophic for a majority of households (shock-heavy), and that this scenario ordering — not the precise magnitude — is the paper's most robust finding, since it survives both the wage/shock-frequency sensitivity analysis (Section 4.5) and the Ghana-empirical recalibration exercise (Section 4.6). These results matter because Ghana's health-financing policy choices over the next two to three decades will determine which of these trajectories Generation Z actually experiences, and the gap between the most and least favorable scenarios is large enough to represent a qualitative difference in household financial security rather than a marginal one. What follows in Sections 5.1–5.3 explains the structural mechanisms behind this divergence, its generational implications, and the specific policy levers the model identifies as capable of altering the outcome.

The scenario divergence documented in Section 4 is driven by a small number of structural mechanisms already visible in Ghana's recent economic history rather than by exotic or speculative assumptions. First, currency exposure in health financing functions as an involuntary, regressive tax: because pharmaceuticals, diagnostic equipment, and hospital consumables are substantially imported and dollar-priced, every episode of cedi depreciation raises health costs independent of domestic wage or productivity conditions — the mechanism behind the 12.8 percent single-year fall in dollar-denominated per-capita health expenditure observed between 2021 and 2022 (Knoema 2025). Second, NHIS coverage in Ghana has historically expanded in breadth (enrollment) faster than in depth (the share of an episode of care actually reimbursed), leaving even insured households exposed to OOP payments for medicines and specialist care (Okoroh and Essoun 2018). Third, government health financing has persistently fallen short of the Abuja Declaration commitment, with real increases in health spending since 2015 driven mostly by wage-bill growth rather than expanded financial risk protection (World Bank 2023) — meaning the financing envelope available to deepen NHIS coverage has not kept pace with population or cost growth.

A gap worth confronting directly: the empirical GLSS7 baseline computed in Section 4.1 (1.08 percent catastrophic-expenditure incidence at a 10-percent threshold) sits well below the 8-percent annual catastrophic-event probability used to calibrate the primary simulation's status-quo scenario. Three factors plausibly explain this gap rather than indicating an error in either estimate. First, the simulation's parameter was deliberately blended from wider Sub-Saharan African literature (12.9–20.4 percent regional CHE incidence) precisely because Ghana-specific, forward-looking parameters were not available prior to this paper's own empirical analysis — the GLSS7 figure computed here is more Ghana-specific but reflects a single cross-section (2016/17) rather than the full range of household experience an individual accumulates over the primary simulation's 25-year working-life horizon, during which the probability of experiencing at least one catastrophic event compounds substantially above any single year's incidence. Second, consumption-diary-style surveys such as GLSS are documented to underreport health spending relative to purpose-built health-expenditure surveys, since health is one module among many rather than the survey's focus. Third, GLSS7 predates several years of subsequent healthcare cost

inflation and currency volatility (including the 2022 crisis), meaning a 2016/17 empirical baseline likely understates catastrophic-expenditure risk under current conditions. Rather than treating this gap as unresolved, Section 4.6 directly tests its consequences: we back-solve and re-run the primary and extended simulations under a Ghana-empirical calibration anchored to the 1.08 percent estimate, finding that the paper's central scenario ordering is preserved, though absolute projected burden is somewhat lower under this alternative than under the original literature-blended calibration.

5.2 Generational implications for Gen Z

Three features of Ghana's economic structure make the generational stakes of this trajectory distinct from those facing the cohort currently at the peak-risk age. First, informal-sector employment — which structurally weakens automatic NHIS premium payment and consistent enrollment — is expected to remain the modal form of labor-market participation for Ghanaian Gen Z, given current formalization rates. Second, Ghana's epidemiological profile is shifting toward non-communicable diseases (hypertension, diabetes, cardiovascular disease, cancers), which generate longer-horizon, repeat-cost treatment needs that OOP-heavy financing systems handle particularly poorly compared to acute, one-off conditions. Third, Gen Z households are likely to face simultaneous caregiving demands for aging parents and their own children — a "sandwich generation" dynamic that compounds household-level financial exposure precisely as the affordability gap identified in Section 4 is widest under the status-quo and shock-heavy scenarios.

The epidemiological shift noted above is not a distant projection but an already-measurable feature of Ghana's disease burden. Non-communicable diseases — principally hypertension, diabetes, cardiovascular disease, and stroke — now account for 43–45 percent of all deaths in Ghana (World Health Organization Regional Office for Africa 2024), and diabetes prevalence among Ghanaian adults, estimated at 3.3–8.3 percent, is explicitly projected to continue rising (Lule et al. 2024). Hypertension prevalence based on direct blood-pressure measurement reaches 51.1 percent among adults aged 50 and older — more than three times the 14.2 percent self-reported rate — indicating a large undiagnosed burden that typically becomes costlier once it progresses to an acute cardiovascular event (Minicuci et al. 2014). Consistent with this pattern, a Ghana-specific analysis of catastrophic health expenditure found that having a household member aged 60 or older increased the odds of catastrophic expenditure by a factor of 1.45, second only to household size among the strongest predictors identified ("An Analysis of Household Catastrophic Health Expenditure and Food Insecurity in Ghana" 2025). More speculatively, if Ghana follows the pattern documented elsewhere — ischemic stroke incidence has been rising specifically among adults under 50 in multiple health systems, a trend attributed to earlier onset of the same risk-factor cluster (obesity, hypertension, dyslipidemia) driving the older-age NCD burden (Zhang et al. 2020) — the age of onset for Ghana's own NCD burden may compress downward over the coming decades as well, though Ghana-specific evidence on this trend was not identified in the course of this research and is flagged here as a hypothesis rather than an established finding.

This shift bears directly on the age-stratified empirical pattern documented for GLSS7 households in this paper: mean OOP burden was found to be roughly flat across working-age households

before rising materially in the 60–69 and 70-plus age groups. That pattern reflects the age-of-onset distribution for chronic disease as it stood in 2016/17. If early-onset non-communicable disease risk continues to rise — as documented globally and, for diabetes specifically, within Ghana — the age at which Ghanaian households reach peak catastrophic-expenditure risk may shift materially earlier for the Gen Z cohort than it did for the cohort measured here — a mechanism distinct from, and additive to, the currency- and wage-driven affordability gap modeled in Section 4. Where the simulation's scenarios capture a widening gap between health-cost growth and wage growth at a fixed age profile, an earlier age-of-onset shift would mean Gen Z households reach that gap sooner in their working lives, compounding financial exposure at an age when income and savings buffers are typically at their thinnest. This is presented as a plausible additional mechanism warranting dedicated future research using age-stratified longitudinal Ghanaian health data, rather than as a quantified component of the simulation itself.

A further dimension raised in the course of this research, informed by clinical consultation, is the role of literacy and health literacy specifically. The GLSS7 data analyzed here show a modest but directionally consistent gap: households headed by a literate individual report a weighted mean OOP burden of 0.74 percent of expenditure, compared to 0.84 percent among households headed by an illiterate individual (n = 12,384 and 1,598 respectively). This is consistent with a wider Ghanaian literature linking education and health literacy to financial exposure — a study of catastrophic expenditure among long-bone-fracture patients across four Ghanaian teaching hospitals found lower catastrophic-expenditure rates among more educated patients (Buunaaim et al. 2025), while a study of malaria treatment costs found a more complex relationship, in which higher caregiver health literacy was associated with higher immediate treatment costs, plausibly reflecting more appropriate healthcare-seeking rather than cost avoidance, with the authors noting that lower health literacy may instead impose costs later through delayed care or reinfection (Ofori Boateng et al. 2025). To test directly whether illiteracy's effect on OOP burden compounds with age, as clinically hypothesized, we examined the literacy gap in OOP burden separately within each age group. The gap did not show a consistent widening pattern across age groups in this cross-sectional sample (ranging from a small negative gap at ages 50–59 to a larger positive gap at ages 70 and older, with no monotonic trend in between); several age-by-literacy cells contain few observations, limiting statistical power to detect a compounding effect even if one exists. Testing this hypothesis properly would require panel data tracking the same households' health spending over time, which a single cross-sectional survey round cannot provide — a further motivation for the AHIES panel validation proposed as future work in Section 7.

5.3 Policy levers

The scenario comparison in Section 4 is, by construction, also a policy comparison: the difference between the status-quo and financing-reform trajectories represents the approximate scale of what closing Ghana's Abuja Declaration financing gap could achieve for generational healthcare affordability. Four levers, consistent with the parameters varied in the financing-reform scenario, are implicated directly by the model: (i) raising government health expenditure toward the 15-percent-of-budget Abuja target, with new financing directed at NHIS reimbursement depth rather than only administrative or wage-bill growth; (ii) strengthening primary and preventive care to reduce the incidence of catastrophic-tier health events, rather than only subsidizing their cost

after they occur; (iii) reducing currency exposure in health-input costs through local pharmaceutical and medical-equipment manufacturing capacity; and (iv) extending consistent NHIS premium mechanisms to informal-sector workers, for whom coverage lapse — not lack of enrollment per se — is often the proximate cause of exposure.

6. Limitations

Beyond the modeling limitations noted in Section 3.6, several caveats bear directly on interpretation. First, the simulation uses a single representative cohort and does not model heterogeneity in occupation, region, or pre-existing wealth beyond what is implicitly captured through the stochastic income and cost processes; the regressive incidence patterns documented in the cross-sectional literature (Ataguba and Kabaniha 2022) suggest that poorer sub-populations within the simulated distribution likely face materially worse outcomes than the aggregate statistics reported here. Second, the model treats government policy parameters (e.g., Abuja target compliance) as exogenously fixed for the full horizon within each scenario, whereas real policy trajectories are likely to shift over 25 years in ways this design cannot capture. Third, the catastrophic-event probability and severity parameters are blended from studies using different survey instruments, thresholds, and time periods, introducing calibration uncertainty that the reported percentile ranges do not fully capture. Finally, the model does not incorporate potential future shocks with no clear historical precedent — a pandemic-scale health-system shock, for instance, or a structural change in NHIS financing architecture — and should be read as a projection of currently observable trends rather than a full risk envelope.

Generation Z is used throughout this paper as an analytical cohort — a shared birth-year window used to anchor the simulation's age trajectory — not as a claim that Ghanaian Gen Z is a homogeneous population facing a single, shared healthcare-affordability outcome. The model does not represent, and this paper does not claim to represent, differences across the cohort in income, employment status, occupation, wealth, region, household size, insurance coverage, or broader socioeconomic status; any of these could plausibly shift an individual household's trajectory well above or below the reported scenario means, in the same direction the regressive incidence patterns documented in Sections 2 and 4.1 would predict. Readers should not infer that all Ghanaian Generation Z individuals, or even a representative Ghanaian Gen Z household, will experience the specific trajectory reported here; the simulation instead characterizes a central tendency and its uncertainty for a stylized representative household under transparent assumptions.

To summarize the paper's evidentiary and methodological limitations in one place: (i) model assumptions throughout are those of a stylized stress test, not a structural or general-equilibrium forecast (Section 3.6); (ii) several parameters carry meaningful calibration uncertainty; this is addressed by both a one-at-a-time sensitivity analysis (Section 4.5, Table 5) and a probabilistic analysis jointly resampling all seven uncertain parameters (Section 4.5, Tables 8, Figure 12), which together indicate the paper's qualitative conclusions are reasonably robust but that precise point estimates carry a wide plausible range (12.3–52.4 percent, 10th–90th percentile, for the 2051 status-quo outcome); (iii) the empirical CHE incidence computed from GLSS7 (1.08 percent) and the simulation's catastrophic-event hazard probability (8 percent, or 1.08 percent under the Ghana-empirical calibration) are conceptually distinct quantities whose relationship is discussed at length in Sections 3.3, 4.6, and 5.1, and readers should not treat either figure as a direct estimate of the other; (iv) the simulated cohort is treated as homogeneous in the specific respects listed above; (v) income and healthcare-cost growth assumptions are extrapolated from historical Ghanaian series and may not hold if either process changes structurally over the projection horizon; (vi) the 25- and 35-year projection horizons carry substantial compounding uncertainty that widens with time, and the extended

(2061) results in particular should be read with more caution than the primary (2051) results, consistent with their separate reporting status (Section 3.2); (vii) the model cannot anticipate future Ghanaian health-financing policy changes and treats each scenario's policy parameters as fixed for the full horizon; and (viii) the debt-trigger liquidity-buffer and threshold assumptions remain author calibrations not drawn from a Ghana-specific household-savings survey, though a sensitivity sweep across a 4×4 grid of plausible buffer and threshold values (Section 3.6, Table 7) shows the qualitative finding — that a majority of simulated households experience at least one debt-triggering event by age 50 under the status-quo scenario — holds across every combination tested, even though the precise probability (51.7–88.7 percent) is sensitive to these choices. Taken together, these limitations mean the paper's numerical projections should be read as illustrating the direction, approximate scale, and relative ordering of generational healthcare-affordability risk under transparent assumptions, not as point forecasts, and the discussion above is intended to make the boundaries of that claim explicit rather than to understate the paper's contribution.

7. Conclusion

This paper has proposed a forward-projecting, cohort-specific complement to Ghana's largely retrospective catastrophic-health-expenditure literature. Calibrating a stochastic stress-test simulation to Ghana's own historical wage, inflation, and health-financing data, our primary simulation (2026–2051, approximately age 25–50) finds that continuation of current trends would bring Ghana's Generation Z cohort to a mean out-of-pocket healthcare burden of 19.6 percent of income by age 50 — comparable to, but not dramatically worse than, the burden documented for today's cohort at the same age — with a materially worse outcome for the upper tail of the distribution, where a substantial minority of households would already be in catastrophic-expenditure territory. Under a scenario of repeated currency and inflation shocks, consistent with Ghana's post-2000 economic history, mean burden more than triples the catastrophic-expenditure threshold by 2051. Under a financing-reform scenario in which government health spending reaches the Abuja Declaration target, burden remains flat or falls. A separately reported extended simulation (2026–2061, approximately age 25–60; Section 4.4) indicates this divergence continues to widen through age 60, though as an extension beyond the paper's original design horizon this finding carries a different evidentiary status than the primary, age-50-anchored results. The generational stakes of Ghana's health-financing policy choices, in other words, are not marginal: the gap between the paper's most and least favorable scenarios represents, for a single cohort, the difference between healthcare remaining a manageable household expense and healthcare becoming Ghana's Generation Z's dominant source of financial precarity.

Beyond the simulation itself, this paper contributes a directly computed, GLSS7-based empirical baseline (Section 4.1) that both corroborates the order of magnitude of prior GLSS7-based catastrophic-expenditure findings (Sataru, Twumasi-Ankrah, and Seddoh 2022) and identifies a meaningful gap between Ghana-specific single-year incidence and the wider regional literature used to calibrate the primary simulation's scenarios (Section 3.3), a gap we directly reconcile in Section 4.6 by re-running both the primary and extended simulations under a Ghana-empirical calibration. The most direct remaining extension of this work is recalibrating the simulation's catastrophic-event parameters against newer AHIES (2022–2024) microdata rather than the single 2016/17 GLSS7 cross-section used here, and, where the AHIES panel structure permits,

validating the simulation's near-term trajectory against observed year-over-year change in household OOP burden rather than against cross-sectional benchmarks alone.

8. Data and Code Availability

The following materials support independent verification and reproduction of this paper's results, consistent with the reproducibility goal set out in Section 3.2: (i) the primary and extended simulation code for all three scenarios (status quo, financing reform, shock-heavy) at both the T = 25 (2026–2051) and T = 35 (2026–2061) horizons; (ii) the GLSS7 extraction and cleaning pipeline described in Section 3.4, including the household-identification, weighting, missing-value, and zero-expenditure handling logic set out there; (iii) the full mathematical specification of the model given in Section 3.2's eleven numbered components; (iv) the calibrated parameter table (Table 1), including the Ghana-specific/external/author-calibrated classification introduced in Section 3.1; (v) the fixed random seed(s) used to generate the reported figures and tables, so the exact reported values — not merely their approximate distribution — can be reproduced; (vi) the one-at-a-time sensitivity-analysis code underlying Table 5 and Figure 11; and (vii) the underlying data for every reported figure and table. Code is implemented in Python 3 (NumPy, pandas, matplotlib) and organized so the primary simulation, extended simulation, GLSS7 extraction, and sensitivity analysis can each be run independently.

Data availability statement. GLSS7 (2016/17) household microdata are held by the Ghana Statistical Service and are available to researchers on application through GSS's standard microdata-access procedure; the underlying microdata cannot be redistributed directly by the author. This paper's own extraction pipeline, and the household-level analysis output it produces, are provided as supplementary material so that a researcher who separately obtains GLSS7 access from GSS can reproduce every statistic reported in Section 4.1 and Table 3 exactly. All other model inputs (Ghana's minimum-wage series, inflation series, and health-financing shares) are drawn from the publicly available sources cited in Table 1 and the References section and require no restricted access.

Code availability statement. All simulation, extraction, and sensitivity-analysis code described above is made available alongside this manuscript as supplementary material, or on request from the corresponding author, subject to the GLSS7 microdata-access restriction described above.

AI Use Disclosure

The author used an AI assistant (Claude, Anthropic) to support literature search and synthesis, drafting of manuscript text, development of the simulation and GLSS7 data-extraction code described in Sections 3.2–3.4, and generation of the figures presented throughout this paper. The AI-generated extraction pipeline was iteratively tested against synthetic data structured to match GLSS7's known file conventions before being applied to the real microdata, and its output was independently reviewed against the underlying data at each stage, including verification of household-count totals, value-label confirmation for categorical variables, and cross-checks against the peer-reviewed literature cited throughout. All modeling decisions, parameter interpretations, scenario definitions, and conclusions are the author's own.

References

Akazili, James, John Ele-Ojo Ataguba, Edmund Wedam Kanmiki, John Gyapong, Osman Sankoh, Abraham Oduro, and Di McIntyre. 2017. “Assessing the Impoverishment Effects of Out-of-Pocket Healthcare Payments Prior to the Uptake of the National Health Insurance Scheme in Ghana.” BMC International Health and Human Rights 17: 13. https://doi.org/10.1186/s12914-017-0121-7.

Akazili, James, Diane McIntyre, Edmund W. Kanmiki, John Gyapong, Abraham Oduro, Osman Sankoh, and John E. Ataguba. 2017. “Assessing the Catastrophic Effects of Out-of-Pocket Healthcare Payments Prior to the Uptake of a Nationwide Health Insurance Scheme in Ghana.” Global Health Action 10 (1): 1289735. https://doi.org/10.1080/16549716.2017.1289735.

“An Analysis of Household Catastrophic Health Expenditure and Food Insecurity in Ghana.” 2025. medRxiv, preprint. https://www.medrxiv.org/content/10.1101/2025.03.31.25324988.full.pdf.

Ataguba, John E., and Grace A. Kabaniha. 2022. “Characterizing Key Misconceptions of Equity in Health Financing for Universal Health Coverage.” Health Policy and Planning 37 (7): 928–931. https://doi.org/10.1093/heapol/czac041.

Bloomberg. 2022. “Ghana Inflation Exceeds 37% as Data Compilation Is Changed.” October 12, 2022. https://www.bloomberg.com/news/articles/2022-10-12/ghana-annual-inflation-at-37-2-as-data-compilation-is-changed.

Buunaaim, Alexis D. B., Michel Adurayi Amenah, Dominic Konadu-Yeboah, Richard Baidoo, Amgbo Asare, Wilfred Larbi Addo, and Claude Martin. 2025. “Evaluating Catastrophic Health Expenditures among Patients with Long Bone Fractures in Ghana's Major Teaching Hospitals: A Hospital-Based Analysis.” BMC Health Services Research.

Demirgüç-Kunt, Asli, Leora Klapper, Dorothe Singer, and Saniya Ansar. 2022. The Global Findex Database 2021: Financial Inclusion, Digital Payments, and Resilience in the Age of COVID-19. Washington, DC: World Bank.

Fair Wages and Salaries Commission. 2025. “2026 National Daily Minimum Wage.” Accra: Government of Ghana. https://fairwages.gov.gh/wp-content/uploads/2025/11/2026-National-Daily-Minimum-Wage.pdf.

Finex Insights. 2025. “Ghana's Minimum Wage Growth: A Decade of Gradual Gains and Sudden Jumps (2015–2025).” August 19, 2025. https://www.finexinsights.com/post/ghana-s-minimum-wage-growth-a-decade-of-gradual-gains-and-sudden-jumps-2015-2025.

Knoema. 2025. “Ghana Health Expenditure per Capita, 1960–2025.” https://knoema.com/atlas/Ghana/topics/Health/Health-Expenditure/Health-expenditure-per-capita.

Lule, Swaib Abubaker, Sandra Boatemaa Kushitor, Carlos S. Grijalva-Eternod, Kafui Adjaye-Gbewonyo, Olutobi Adekunle Sanuade, Mawuli Komla Kushitor, Lydia Okoibhole, Raphael Awuah, Leonard Baatiema, Irene Akwo Kretchy, Daniel Arhinful, Ama de-Graft Aikins, Kwadwo Koram, and Edward Fottrell. 2024. “The Contextual Awareness, Response and Evaluation (CARE) Diabetes Project: Study Design for a Quantitative Survey of Diabetes Prevalence and Non-Communicable Disease Risk in Ga Mashie, Accra, Ghana.” Global Health Action 17 (1): 2297513. https://doi.org/10.1080/16549716.2023.2297513.

Minicuci, Nadia, Richard B. Biritwum, George Mensah, Alfred E. Yawson, Nirmala Naidoo, Somnath Chatterji, and Paul Kowal. 2014. “Sociodemographic and Socioeconomic Patterns of Chronic Non-Communicable Disease among the Older Adult Population in Ghana.” Global Health Action 7: 21292. https://doi.org/10.3402/gha.v7.21292.

Ofori Boateng, Millicent, Derek Asuman, Nuworza Kugbey, Padmore Adusei Amoah, Peter Agyei-Baffour, and Ulrika Enemark. 2025. “Health Literacy and Household Financial Loss on Malaria Treatment for Children under Five in Ghana: A Patients' Perspective.” International Health 17 (1): 77–83.

Okoroh, Juliet, and Sarpong Essoun. 2018. “Evaluating the Impact of the National Health Insurance Scheme of Ghana on Out-of-Pocket Expenditures: A Systematic Review.” BMC Health Services Research 18: 426. https://doi.org/10.1186/s12913-018-3249-9.

Playroll. 2026. “Minimum Wage in Ghana: Rates, Trends & Compliance.” https://www.playroll.com/minimum-wage/ghana.

Sataru, Fuseini, Kwame Twumasi-Ankrah, and Anthony Seddoh. 2022. “An Analysis of Catastrophic Out-of-Pocket Health Expenditures in Ghana.” Frontiers in Health Services 2. https://doi.org/10.3389/frhs.2022.706216.

Trading Economics. 2026. “Ghana Inflation Rate.” Accessed August 2026. https://tradingeconomics.com/ghana/inflation-cpi.

World Bank. 2023. “Public Health Expenditure for Universal Health Coverage in Ghana.” Washington, DC: World Bank Group. https://documents1.worldbank.org/curated/en/099928302092432757/pdf/IDU169cd1ed015856 14b481be7b1bd94d105e459.pdf.

World Bank. n.d. “Out-of-Pocket Expenditure (% of Current Health Expenditure) – Ghana.” World Bank Open Data, WHO Global Health Expenditure Database. Accessed August 2026. https://data.worldbank.org/indicator/SH.XPD.OOPC.CH.ZS?locations=GH.

World Health Organization Regional Office for Africa. 2024. “WHO Supports Ghana in Developing Landmark 2023 STEPS Report to Address Non-Communicable Disease Risk Factors.” November 20, 2024. https://www.afro.who.int/countries/ghana/news/who-supports-ghana-developing-landmark-2023-steps-report-address-non-communicable-disease-risk.

Zhang, Nan, Lin Zhang, Qiu Wang, Jingwei Zhao, Jia Liu, and Guang Wang. 2020. “Cerebrovascular Risk Factors Associated with Ischemic Stroke in a Young Non-Diabetic and Non-Hypertensive Population: A Retrospective Case-Control Study.” BMC Neurology 20: 424. https://doi.org/10.1186/s12883-020-02005-7.